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PRESENTATION FINAL PROJECT – UNIT 2

EIGENVALUES & EIGENVECTORS IN AI

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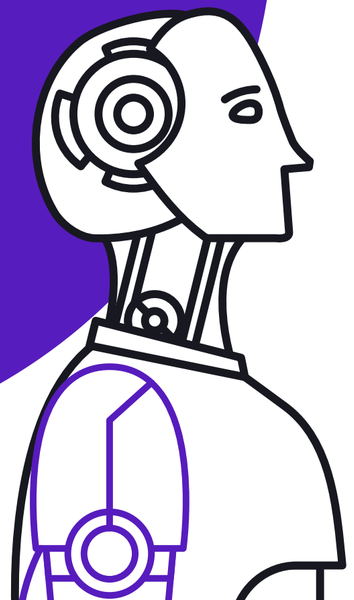

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TOOLS USED

NumPy

pandas

scikit learn
machine learning in Python

matplotlib

CO

kaggle

SymPy

GitHub

PROBLEM STATEMENT

In Artificial Intelligence, we often deal with high-dimensional data.

More dimensions mean:

01 More complexity

02 More noise

03 Higher computational cost

ABOUT THE DATASET

This dataset contains information about Spotify tracks across 125 different music genres,

df.shape( )
(114000,21)

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WHAT IS AN EIGENVALUE?

An eigenvalue (λ) tells us:

How much scaling happens in that direction

Large λ → strong stretching

Small λ → weak effect

WHAT IS AN EIGENVECTOR?

$$A\vec{x} = \lambda\vec{x}$$

$n \times n$
Matrix

Eigenvector

Eigenvalue

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WHY THIS MATTERS IN DATA

01

IS THERE ANY SPREAD?

Data spreads differently across directions

02

IS THERE ANY INFO ?

- Some directions contain more information.

03

WHICH PCS MOSTLY CONTAIN NOISE

SOME DIRECTIONS MOSTLY CONTAIN NOISE.

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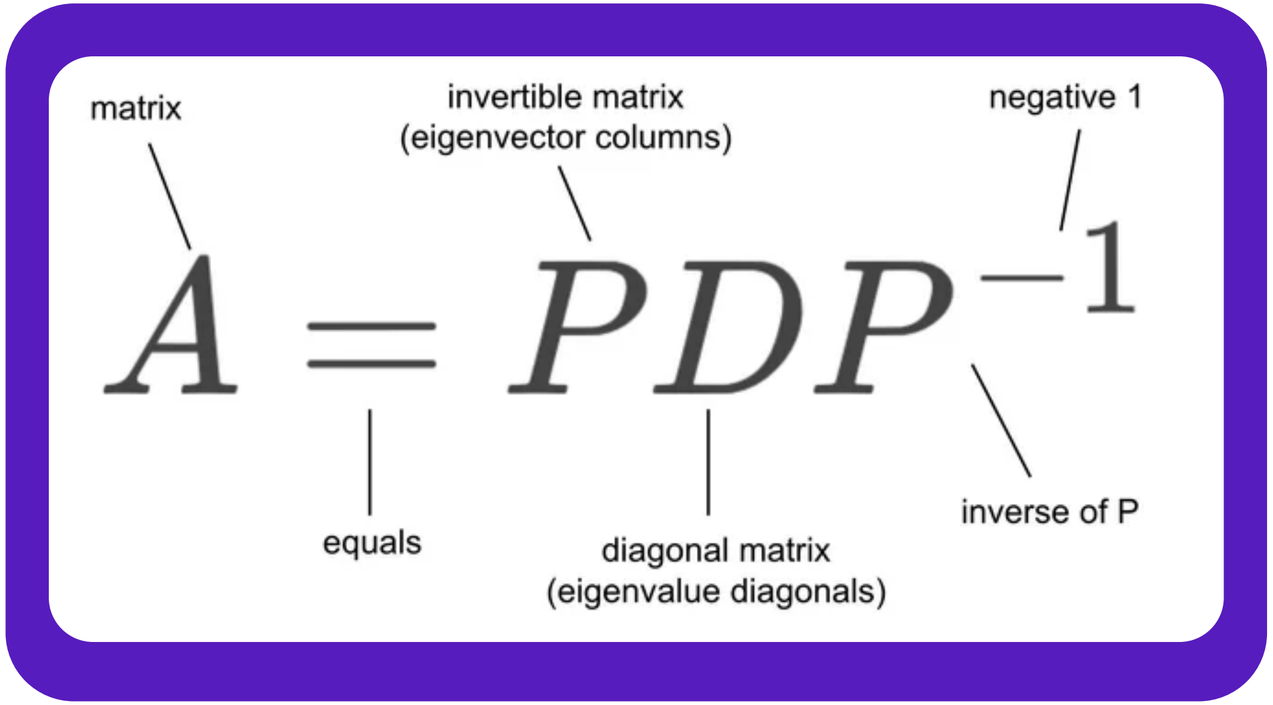
DIAGONALIZATION

Diagonalization breaks a matrix into directions (P), their importance (D), and transforms back to the original space (P^{-1}), making complex data easier to understand."



CONNECTION TO AI

- Feature reduction
- Image compression
- Face recognition
- Recommender systems

WHAT IS PCA?

Principal Component Analysis (PCA) is:
A technique that reduces dimensionality while preserving maximum variance.
Instead of keeping all original features, PCA creates new ones called:

PRINCIPAL COMPONENTS (PCS)

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PRINCIPAL DIRECTIONS

Each Principal Component:

- Is an eigenvector of the covariance matrix
- Points to a direction of maximum variance

PC1 → MAXIMUM VARIANCE
PC2 → SECOND MAXIMUM
PC3 → THIRD MAXIMUM

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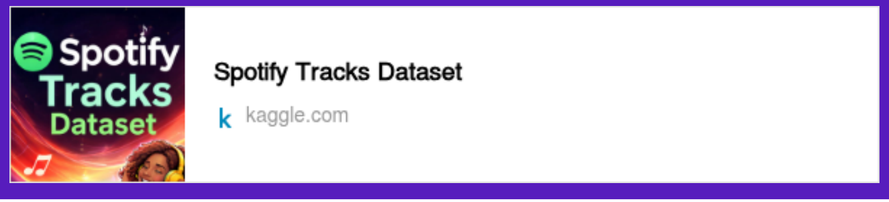
REFERENCES

Dataset



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CODE

GitHub

TUWAIQ-LINEAR-ALGEBRA

THANK YOU

NUMPY PROJECT